



BTEBE202 /
BSC202 / 204T

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School of Engineering & Technology

B.Tech. II Sem : Biomedical / Biomedical+MBA / Civil /

Electrical / Electronics / Mechanical / Biotechnology /

**Biotechnology + MBA / Chemical / Computer Science / Food Technology /
CSE (AIML) / CSE (Cyber Security)**

II SEMESTER END EXAMINATION: MAY 2022

ENGINEERING CHEMISTRY - II

Max. Marks: 70

Time: 3 Hrs.

INSTRUCTIONS

Attempt all units. Internal choices are given in each unit.

UNIT-I

- Q.1** (a) Discuss the Proximate and Ultimate Analysis of a fuel. 7
- (b) Comment on "Determination of Calorific Value by Bomb Calorimeter". 7

OR

- (a) Differentiate between LTC and HTC. What is the significance of Calorific value in selection of a fuel? 7
- (b) Write a detailed note on Otto-Hoffmann By-product Coke Oven method. 7

UNIT-II

- Q.2 (a) Comment on Knocking and Octane Number.
(b) Compare Solid, Liquid and Gaseous fuels.

OR

- (a) Discuss Catalytic Cracking and its types.
(b) Write a detailed note on Refining of Petroleum.

UNIT-III

- Q.3 (a) Write short notes on Dry and Wet theories of Corrosion.
(b) Comment on various methods for Corrosion Control and Protection.

OR

Write short notes on:

- (i) Blended Oils
(ii) Cloud and Pour Point
(iii) Flash and Fire Point
(iv) Viscosity Index

5

3

3

3

UNIT-IV

Define Crystallization. Discuss Distillation and its different types.

7

- (b) Discuss the Phase Rule on One Component water system.

7

OR

- (a) Define Adsorption Chromatography. Discuss Thin Layer Chromatography and applications of Chromatography.

7

- (b) Comment on - "2 Phase Ag-Pb System".

7

UNIT-V

- Q.5 (a) Derive the Integrated Rate Equation for a First Order Kinetics. Define Rate Constant.

7

- (b) Discuss the factors influencing rate of a reaction.

7

OR

- (a) Compare the 'Collision Theory' and 'Transition State Theory'.

8

- (b) Define Activation Energy. Compare the various aspects for Zero, First and 2nd Order Kinetics.

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